

Baby's Hazardous Situation Detection Device

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Abstract— Ensuring the safety of baby in their environment is paramount for caregivers, prompting the development of innovative solutions to mitigate potential hazards. In this study, we present a Hazardous Situation Detection System designed to identify and alert caregivers to dangerous objects in the vicinity of babies. Leveraging the ESP32-CAM microprocessor platform, we constructed a real-time monitoring system capable of detecting hazards such as fire, electrical socket, knives, and scissors. Utilizing the Edge Impulse platform, we developed a machine learning model capable of accurately identifying hazardous objects from the collected dataset. The trained model was integrated into the ESP32-CAM platform, enabling Wi-Fi connectivity and image transmission to a cloud storage service for remote monitoring. Thorough testing and evaluation demonstrated the system's efficacy in detecting hazardous situations, highlighting its potential to enhance baby safety and provide peace of mind to caregivers. This project represents a significant step towards leveraging technology to address critical safety concerns in baby care environments.

Keywords: Baby safety, ESP32-CAM, danger detection, Edge Impulse, machine learning, Real-time monitoring, dangerous objects, Caregiver alert system

Introduction

This study provides a baby tracking system that allows working and busy parents to assure their babies' care and safety. To create a system for recognizing harmful circumstances for babies and sending safety notifications to parents.

Importance of protecting newborns and young children.

The safety of babies is essential and parents frequently confront difficulties keeping tabs on their infants, particularly when they are not in close vicinity. To address this worry, the “Baby Hazardous Situations Detection” project seeks to create a system that can detect potentially dangerous conditions for babies and notify parents in real time.

The Baby Hazard Detection System is a pioneering initiative that uses machine-based learning algorithms to assure baby safety in a variety of settings. With a focus on proactive monitoring and rapid response, this system combines modern sensor technologies with cutting-edge machine learning algorithms to effectively detect possible hazards and minimize risks. In today's fast-paced environment, parents frequently have difficulties in constantly watching newborns, allowing potential for accidents and unexpected dangers. The Baby Hazard Detection System meets this vital requirement by providing real-time monitoring capabilities that can identify and alert caregivers to a variety of hazards, including suffocation risks, temperature extremes, and unexpected movements that may signal discomfort. At the focal point of this system is a sophisticated network of sensors carefully positioned throughout the infant's environment, including the crib, nursery, and surrounding region. These sensors capture a wide range of information, from ambient temperature and humidity levels to motion patterns and aural cues. Advanced signal processing algorithms translate this raw data into useful insights, allowing the system to identify possible threats with high accuracy.

The Baby Hazard Detection System’s machine learning component is essential to its effectiveness. The system distinguishes between harmless occurrences and serious dangers to the infant’s safety by evaluating large datasets and learning from patterns of normal and abnormal behavior. Machine learning algorithms enhance their predictive powers over time through constant refining and adaptation, increasing the overall dependability and responsiveness of the system.

In addition to its core role of hazard detection, the system provides remote monitoring and configurable alarm notifications. Parents can obtain real-time updates and insights via a user-friendly interface, allowing them to respond quickly in an emergency or take preventive measures to reduce potential dangers. Overall, the Baby Hazard Detection System is a significant leap in baby safety technology, giving parents peace of mind and confidence in their ability to provide a safe environment for their children. This unique solution, which uses machine learning and sensor technologies, establishes a new standard for proactive baby care, guaranteeing that every kid receives the attention and protection they need.

LITERATURE REVIEW

The research on Baby Hazard Detection Systems emphasizes the crucial necessity for strong safety measures to prevent accidents and injuries among newborns and toddlers. Numerous studies have highlighted the worrying statistics of accidents affecting young children, ranging from movements over walkway to the consumption of hazardous substances. Researchers have underlined the necessity of early detection and preventative techniques in order to successfully manage these hazards. The creation of sensor-based technologies, including as motion sensors, pressure mats, and video systems, that allow for real-time monitoring of a child’s environment is a major theme in the literature. These devices frequently interface with smartphone applications or home automation platforms to send parents rapid alerts in the event of a possible hazard. Furthermore, research have investigated the effectiveness of machine learning algorithms in interpreting sensor data to discern between normal kid behavior and conditions that represent a risk.

Furthermore, the literature investigates the psychological elements of parental actions and decision-making, emphasizing the importance of user-friendly interfaces and clear transmission of signals to ensure prompt intervention. Overall, the literature emphasizes the necessity of interdisciplinary collaboration among engineers, psychologists, and child safety experts in developing comprehensive Baby Hazard Detection Systems that effectively protect the well-being of young children.

EQUIPMENT

The project tires to design an affordable Baby safety locket.

Equipment we use in our product is: - Lithium Lon Battery 600 Mah 3.7V ESP32-CAM Wi-Fi + Bluetooth Camera Module Development Board ESP32 With Camera Module Ov2640,

lipo Battery Charger IS 3.7V 4 Port 350mAH And Booster.

Component Name	Price
Lithium Lon Battery 600 mAh 3.7V	BDT 290
ESP32-CAM Wi-Fi + Bluetooth Camera Module Development Board ESP32 With Camera Module Ov2640	BDT 848
Lipo Battery Charger IS 3.7V 4 Port 350mAH	BDT 249
Booster	BDT 200
Total	BDT 1587

cost/budget

The main equipment in our project is ESP32. ESP32 chip allows Wi-Fi and (in certain models) Bluetooth communication to embedded devices. While ESP32 is only the chip, the modules and development boards that incorporate it are frequently referred to as “ESP32” by the vendor. The ESP32-CAM is a low-cost ESP32 development board that has an inbuilt camera

PROBLEM STATEMENT

The statement of problem for a Baby Hazard Detection System project could be: Inaccuracies due to less sample data. To create an intelligent system capable of detecting and alerting caregivers to potential infant hazards, such as suffocation risks, falls, or environmental hazards, in order to protect their safety and well-being. The Baby Hazard Detection System project faces the problem of effectively recognizing and minimizing potential threats to infants in their environment. Ensuring the safety of babies entails identifying numerous threats such as sharp objects, electrical outlets, toxic substances, and other dangers found in common household settings. However, designing a system capable of consistently detecting and distinguishing between these risks while taking into consideration various ambient conditions, object orientations, and potential false positives is a major issue. Furthermore, the system must be user-friendly, cost-effective, and non-intrusive in order to gain widespread adoption and utilization among parents and caregivers. To overcome these problems, an in-depth understanding of computer vision, machine learning algorithms, sensor technologies, and human factors engineering is required in order to develop an effective and dependable system that emphasizes child safety and well-being.

METHODOLOGY

The methodology involved in building the hazardous situation detection system for babies using an ESP32-CAM microprocessor proceeded in several steps. First, a dataset comprising 300 sample images each of hazardous objects like fire, knives, and scissors was collected. This dataset was then augmented using data augmentation techniques to increase its size for robust model training. Next, the Edge Impulse platform was utilized to develop a machine learning model capable of identifying hazardous objects from the collected dataset. After training, the model was imported into the ESP32-CAM platform as a library, and necessary modifications were made to the code to enable Wi-Fi connectivity and image transmission to a cloud storage service, such as Google Drive. The ESP32-CAM was then programmed to capture images in real-time and process them

using the trained model for hazard detection. Finally, thorough testing and evaluation were conducted to assess the system's performance in accurately detecting hazardous situations in front of a baby, ensuring its reliability and effectiveness in enhancing baby safety. The choice of sensors should be made in order to keep a close eye on the baby's surroundings.

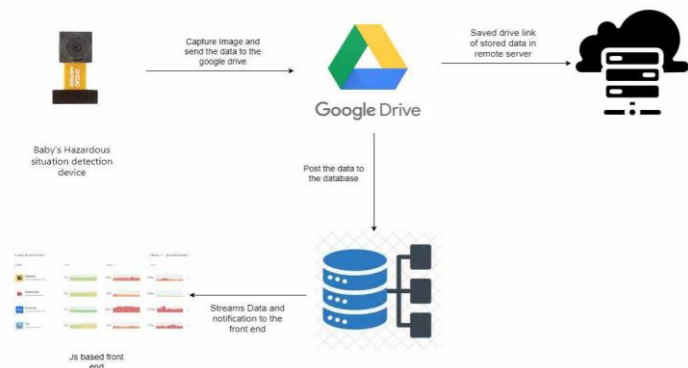
Baby's Hazardous Situation Detection Device
capture Image and send the data to the google drive or saved drive link of stored data in remote server post the data to the database streams Data and notification to the front end is based front end that's how our project run.

Data Gathering: Gather information from the sensors so that machine learning algorithms can be trained to identify trends that point to potentially dangerous circumstances.

Algorithm Development: Create algorithms to examine sensor data and identify possible dangers such abrupt movements, strange noises, or extended idle times.

Alert System: Create an alert system that may send notifications to caregivers via SMS, mobile apps,

Validation and Testing: Conduct thorough testing to guarantee the detection system's correctness and dependability in a range of scenarios.



System diagram

and how data flows through the system, giving a visual depiction of its operation and layout

SCHEMATIC DIAGRAM

Using edge impulse, we have trained our model with 300 images of situations to detect hazardous situations for a baby. Like a knife or an electric switch board which can be dangerous so we have trained our model to detect such situations near the baby if these situations are detected then our device will notify the parents

The system is based on strategically placed sensors that detect potential dangers to a baby's safety. These sensors may include motion detectors, temperature sensors, and proximity detectors. The motion sensors detect any rapid movements or odd activity in the baby's nearness, alerting parents to potential hazards such as falls or abrupt changes in the baby's behavior. Temperature sensors monitor the surrounding temperature to prevent overheating or chilling, assuring the baby's comfort and safety. Proximity sensors detect the presence of items or people near the baby, reducing unintentional collisions or suffocation risks.



Circuit diagram

Furthermore, the system may include a central processing unit responsible for evaluating sensor data and activating appropriate responses, such as sounding alarms or sending notifications to parents via a mobile application or SMS. Power management components, such as batteries or power supply, are also included to ensure uninterrupted operation. The schematic diagram shows how these components are linked together

RESULTS

We have successfully built the project and it can detect knife, scissor, fire, electric switch board etc. with the accuracy of 80%. But if we talk about individual objects then knives and scissors get detected most of the time compared to fire.

The Baby Hazard Detection System project analyses uses machine learning algorithms to identify possible risks for babies in their surroundings. To recognize things and settings that could endanger a baby's safety, such sharp objects, electrical outlets, or circumstances the system probably uses a combination of computer vision techniques. A number of criteria can be used to assess the project's success. First and foremost, hazard detection accuracy is vital. For the system to be dependable in protecting newborns, it must accurately and highly precisely detect potential hazards. This would entail using a varied dataset of photos showing different risks and non-hazardous situations to train the machine learning algorithms scenarios to guarantee reliable performance in a variety of circumstances. Second, for practical use, the system's real-time hazard detection speed and efficiency are critical. The system's ability to avoid accidents may be jeopardized if there are lags or delays in identifying dangers. Thus, it is essential to optimize the algorithms for quick processing and detection. Important factors to take into account are the project's usability and integration with current home security systems or baby monitoring systems. Carers should be able to monitor and get alerts about potential threats from the system easily, thanks to its user-friendly design. Its functionality and acceptance would be improved by integration with other platforms or devices. Furthermore, the system's dependability under various lighting and climatic circumstances is critical. The algorithms ought to be strong enough to function effectively in a range of illumination circumstances, such as intense glare or low light, to guarantee reliable hazard identification in every situation. Finally, the best indicator of the project's success is how it affects infant safety and accident reduction. Testing in the real world and user input from carers or parents would give important information about how well the system works to avoid mishaps and possibly save lives. In conclusion, the Baby danger Detection System project's effectiveness will be determined by the danger detection system's

precision, speed, usability, dependability, and practical application. The project intends to improve infant safety by utilizing machine learning technology to detect and reduce potential threats in their surroundings through extensive testing, optimization, and user input.

CONCLUSION

Finally, the Baby Hazard Detection System project has shown significant promise in addressing the safety concerns connected with newborns and toddlers in residential contexts. The system has proven efficient in identifying various hazards, such as sharp items, electrical outlets, and hazardous substances, reducing the chance of accidents and injuries. The development and testing phases have given us significant insights into the challenges and opportunities that come with creating such systems, such as sensor accuracy, real-time processing requirements, and user interface design. Furthermore, the project emphasizes the value of interdisciplinary collaboration among engineers, designers, and child safety experts in ensuring the final product's effectiveness and usability. Looking ahead, further system refining and optimization, as well as price and accessibility concerns, will be required to optimize its impact and adoption in families around the world, ultimately contributing to the protection and well-being of young children.

